THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 01/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High online return rates exceeding 25% caused by customers misjudging furniture scale, dimensions, and color accuracy in their living spaces prior to purchase.

2. AI RATIONALE & ANALYSIS

"Augmented reality directly solves spatial sizing and color mismatch by projecting photorealistic, true-to-scale 3D models into the customer's actual room before checkout. THEXRA View provides rapid WebAR and native app rendering without heavy engineering friction, making it ideal for fast deployment within the client's aggressive timeline."

3. PROPOSED SOLUTION CONCEPT

A lightweight web- and mobile-ready augmented reality viewer enabling consumers to place true-to-scale 3D furniture models into their living spaces with dynamic lighting and surface anchoring.

IMPLEMENTATION APPROACH:

1. Phase 1: 3D model optimization and lightweight USDZ/glTF asset pipeline establishment
2. Phase 2: Storefront integration of THEXRA View WebAR on key product detail pages
3. Phase 3: User acceptance testing, cross-browser optimization, and production roll-out

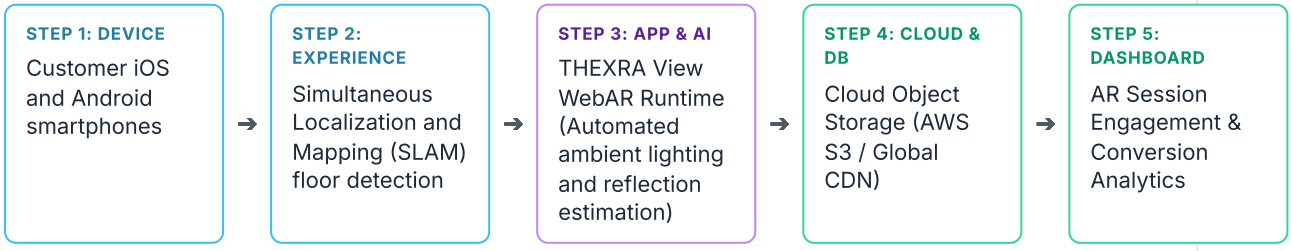
DETAILED IMPLEMENTATION PLAN:

- **Discovery** (1 week): Product catalog asset audit • Technical integration specification
- **Design** (1.5 weeks): WebAR button placement and modal UI flow • PBR material and scale accuracy standards
- **Prototype** (1.5 weeks): Working WebAR prototype for top 5 furniture SKUs • Spatial accuracy validation report
- **Development** (3 weeks): THEXRA View embedded into staging storefront • Batch ingestion of 3D furniture models
- **Testing** (1.5 weeks): Cross-device iOS and Android QA matrix • Lighting and plane-detection benchmark sign-off
- **Deployment** (1.5 weeks): Production launch on live e-commerce site • Real-time analytics and telemetry tracking verification
- **Support** (Ongoing): SLA maintenance and new SKU pipeline support • Monthly return rate and conversion impact analysis

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

AR — THEXRA View

AUTOMATED SYSTEM FLOW DIAGRAM:



> **1. Overview:** Cloud-hosted 3D asset repository integrated with THEXRA View WebAR engine, rendering true-to-scale furniture models via standard mobile web browsers and apps. Instant surface plane detection and realistic PBR shaders ensure exact spatial footprint and accurate material visualization.
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> **2. User Journey:** Customer selects a furniture item on the product page → Taps 'View in Your Space' → Camera scans floor plane → 1:1 scale 3D model anchors in room → Customer verifies size, fit, and color → Confidently adds item to cart.
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> **3. Hardware:** Customer iOS and Android smartphones, LiDAR-compatible mobile devices and tablets
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> **4. Software:** THEXRA View WebAR Runtime, glTF and USDZ 3D Asset Conversion Pipeline, E-commerce Storefront Web Integration SDK
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> **5. AI Components:** Automated ambient lighting and reflection estimation, Surface boundary and edge-detection assistance
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> **6. XR Components:** Simultaneous Localization and Mapping (SLAM) floor detection, Real-time spatial scaling and visual occlusion
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> **7. Backend:** Cloud Object Storage (AWS S3 / Global CDN), RESTful 3D Asset Management API
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> **8. Dashboard:** AR Session Engagement & Conversion Analytics, 3D Asset Management & Ingestion Portal
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> **9. Data Flow:** Product Catalog → Cloud 3D Asset CDN → THEXRA View Engine on Client Device → Real-Time SLAM Display → Session Analytics Telemetry
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> **Alternative Option:** Combination solution — THEXRA Suite (Remote, Make, View)

5. EXPECTED BUSINESS OUTCOMES

Up to 35% reduction in furniture returns, 40% increase in checkout conversion rates, and enhanced buyer spatial confidence.

6. IMPLEMENTATION NEXT STEPS

Audit top 20 highest-return furniture SKUs to convert CAD/3D assets and initiate a rapid THEXRA View WebAR proof of concept.